

Mini Offline UPS



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Most of the systems are powered by AC mains. Uninterrupted power supply systems (UPSes) are there as a back-up to power the systems when mains supply is interrupted due to a power cut. A UPS differs from a standby generator in that it will provide near-instantaneous power by supplying the energy stored in batteries. In an online UPS, batteries are always connected to the inverter, which is always on, so that no transfer switches are necessary when power disruption occurs.

In an offline UPS, the inverter circuit is switched on when mains are not there. UPSes are available off-the-shelf, and one can select the system as per one's requirement, back-up time being one requirement. However, one can construct a UPS of one's own choice. Here is a circuit of an offline UPS, which a hobbyist can make at a reasonable cost. The capacity of the UPS is 350VA, so it can be used for any equipment with a load below 350VA. The UPS can be upgraded to 1kVA by making just a few changes.

Circuit and working

The circuit diagram of the mini offline UPS shown in Fig. 1 has the following four sections:

Section 1: Mains/inverter change-over section

Section 2: Inverter section

Section 3: Battery-status-indicator section

Section 4: Oscillator section

Above-mentioned sections and their interconnections are appropriately marked in the circuit diagram.

Mains/inverter changeover section. The circuit of this section is built around step-down transformer X1 (230V AC primary to 12V-0-12V, 500mA secondary), a 12V DC, 3C/O (changeover) relay (RL1) and a few

other components. 230V AC mains is connected to the circuit through connector CON7. Output of the UPS is available at connector CON8, which is actually a 3-pin socket.

Mains voltage is rectified by diodes D1-D2 (1N4007). The full-wave rectified output is smoothed by capacitor C1. DC voltage so generated is applied at pins 10 and 11 of relay RL1. When mains voltage is there, the relay gets energised to affect the changeover to connect the mains to

the output of the UPS.

The circuit is not on a PCB and has been wired externally. Relay contacts in the circuit diagram are shown in a de-energised state of the relay.

Inverter section. This section comprises transformer X2, npn power transistors 2N3055 (T1 through T8) and power diodes 1N5407 (D3-D4). Transistors, which are eight in number, are connected in two banks. The number of transistors per bank will depend on the required VA rating. The prototype has been made for 350VA rating by using four transistors per bank. The number of transistors required per bank for different capacities are:

550VA – Five

650VA – Six

1000VA – Seven

This circuit is also not on a PCB and has been wired externally. Transistors T1 through T8 have been fitted on the same heat-sink. Mounting of transistors has to be done in such a manner that their base and emitter are not in contact with the heat-sink. The metal body of the transistor is the collector. Collectors should be separated from the heat-sink. This is done by using mica separators between the heat-sink and metal body of the transistor. In short, all three terminals should be separated from the heat-sink. The arrangement is shown in Fig. 2.

Interconnections of transistor terminals, transformer X2, diodes D3-D4, battery-status-indicator section and oscillator section are shown in a combined circuit diagram (Fig. 1). The heat-sink should also be isolated from the UPS box.

Battery-status-indicator section.

This section monitors the state of the battery. It is connected to the battery by CON3-CON4 combination. Con-

PARTS LIST

Semiconductors:

IC1	- 4027 dual JK flip-flop
IC2	- NE555 timer
IC3	- 7805, 5V regulator
T1-T8	- 2N3055 npn power transistor
T9-T10	- BC547 npn transistor
T11-T12	- SK100 pnp transistor
T13-T15	- 2N2222 npn transistor
D1-D2, D5	- 1N4007 rectifier diode
D3-D4	- 1N5407 rectifier diode
LED1-LED3	- 5mm LED
ZD1-ZD2	- 3.6V zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

R1-R2	- 680-ohm
R3-R4	- 150-ohm, 2W
R5-R6, R12	- 3.3-kilo-ohm
R7-R8, R14, R18, R21	- 10-kilo-ohm
R9-R10, R16, R19-R20	- 1-kilo-ohm
R11	- 1.5-kilo-ohm
R13, R17	- 2.2-kilo-ohm
R15	- 22-kilo-ohm
R22	- 4.7-kilo-ohm
VR1	- 10-kilo-ohm preset
VR2, VR3	- 2.2-kilo-ohm preset

Capacitors:

C1, C6	- 220 μ F, 25V electrolytic
C2	- 0.1 μ F ceramic disks
C3	- 0.01 μ F ceramic disk
C4-C5	- 1 μ F, 25V electrolytic
C7	- 330 μ F, 35V electrolytic

Miscellaneous:

X1	- 230V AC primary to 12V-0-12V, 500mA secondary transformer
X2	- 15V-0-15V secondary and 350VA inverter primary transformer
SL, S2	- On/off switch.
RL1	- 12V, 3C/O relay with 6A contact rating
CON1	- 230V AC neon lamp
F1	- 1A fuse
CON1-CON7	- 2-pin connector
CON8	- LED holders (three)
BATT.1	- 6A, 3-pin output socket
	- 12V, 7Ah battery

Test Points

Test point	Details
TP0, TP5	0V, GND
TP1	12V when S2 is closed
TP2	5V
TP3, TP4	2.2V
TP6	12V